

Introduction into Auditory Kernels

The efficient coding hypothesis states that biological sensory systems maximize information transfer to the brain while minimizing neural resources. For the auditory system, a consequence is that incoming signals are encoded sparsely. This was successfully modeled for humans by Smith and Lewicki [1, 2], who showed that sparse coding of natural speech resembles auditory nerve fiber responses. For non-human auditory processing, this phenomenon remains relatively unexplored. In this work, we examine echolocation calls of the bat species *Rhinolophus affinis*.

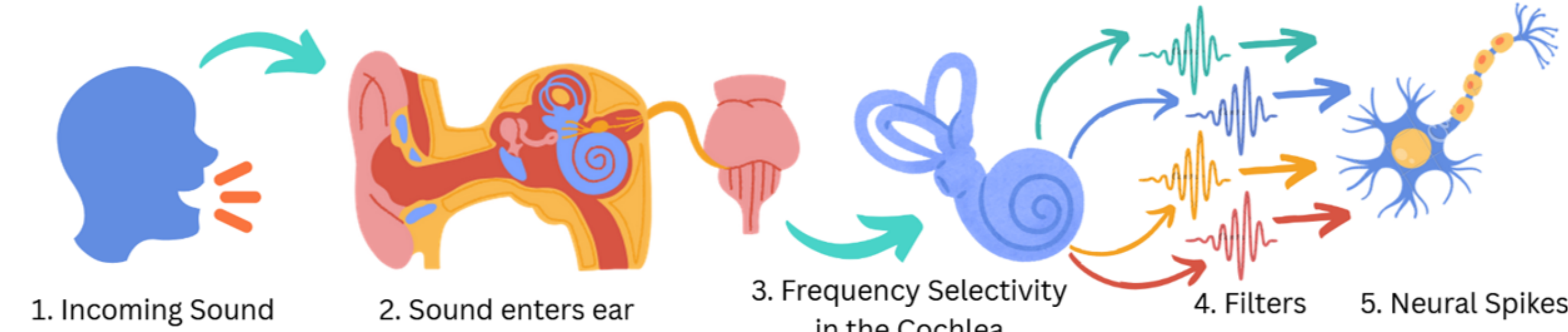


Figure 1. Human auditory processing. Schema of the early auditory system illustrating the efficient coding hypothesis.

Sparse Coding. Decompose signals $x(t)$ into a set of discrete acoustic elements, called auditory kernels, $\phi_{\gamma_k}(t)$, each scaled by $a_k \in \mathbb{R}$ and characterized by a set of parameters $\gamma_k = (\tau_k, \omega_k, s_k) \in \Gamma$, specifying the time shift τ_k , center frequency ω_k , and scale s_k where K is the number of active atoms and Γ the set of all possible atom parameters. The decomposition is given by the following equation:

$$x(t) \approx \sum_{k=1}^K a_k \phi_{\gamma_k}(t), \quad K \ll |\Gamma|$$

$x(t)$ – signal
 $\phi_{\gamma_k}(t)$ – auditory kernel
 $a_k \in \mathbb{R}$ – kernel coefficient
 $\gamma_k = (\tau_k, \omega_k, s_k)$ – kernel parameters
 τ_k – time shift
 ω_k – center frequency
 s_k – scale
 K – number of active kernels
 Γ – set of all kernel parameters

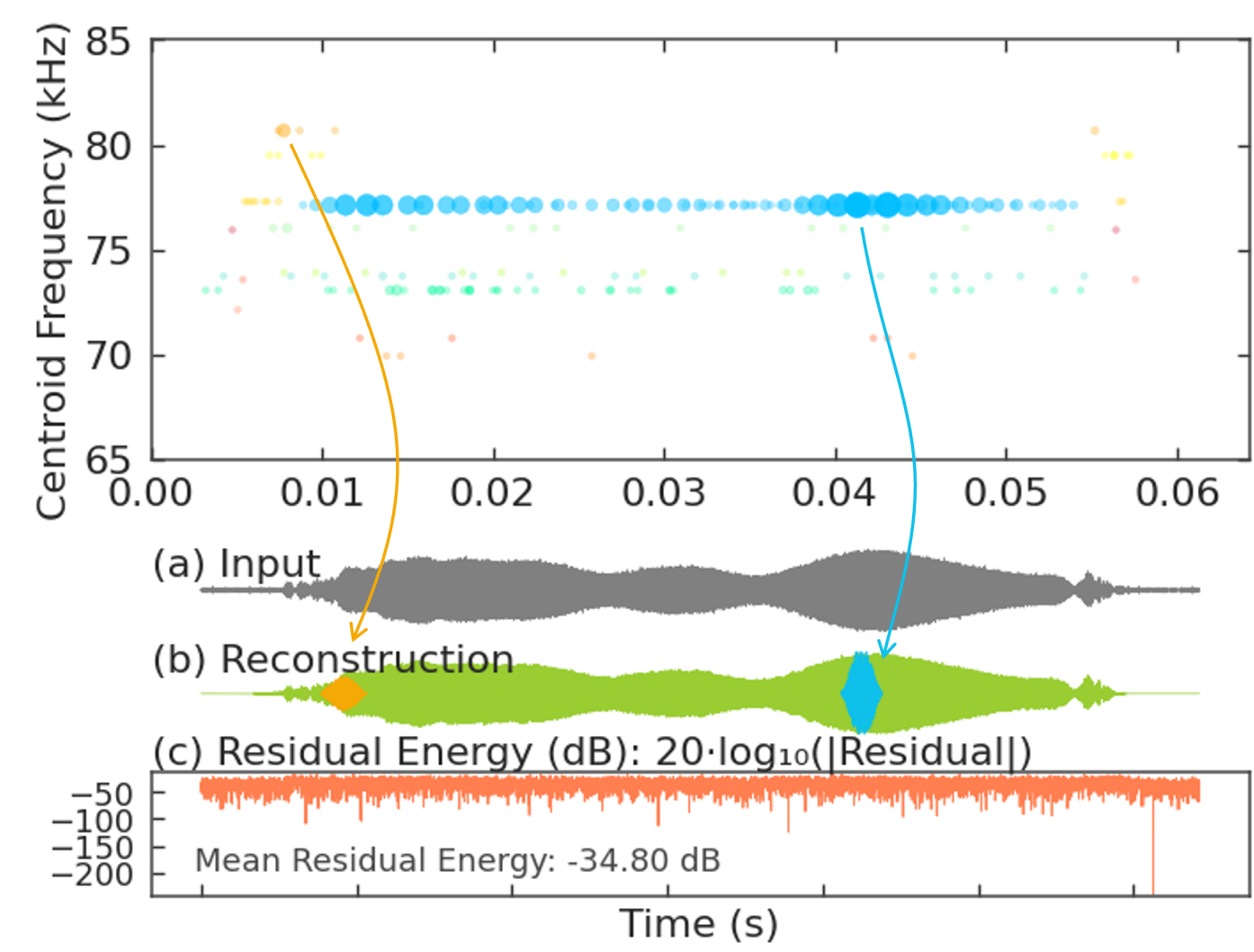


Figure 2. Sparse coding decomposition of a bat echolocation call. Spikes (circles) shown over time and centroid frequency with size proportional to amplitude.

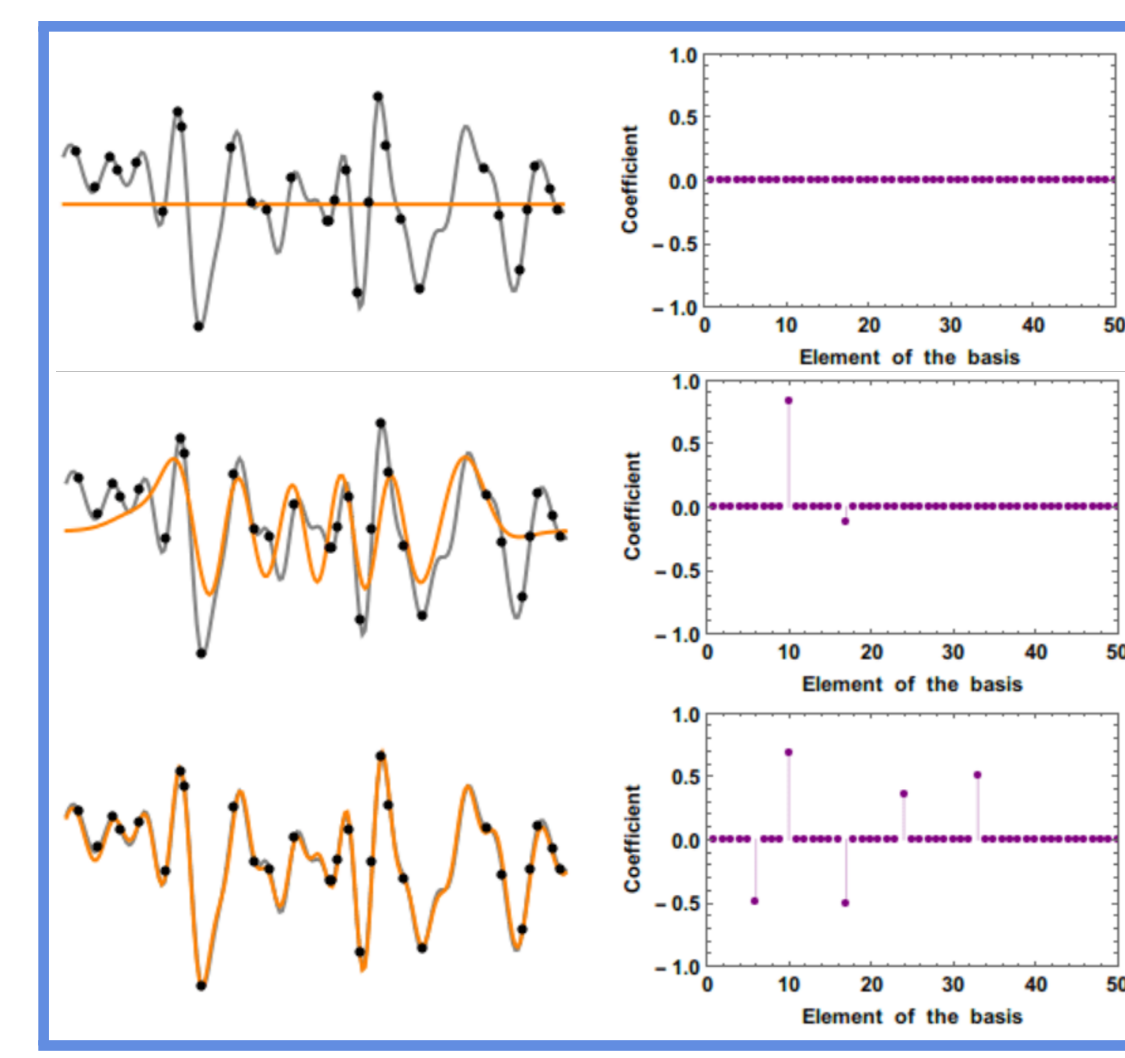


Figure 3. Matching pursuit [3] decomposition. Iterative signal approximation (orange line) with kernel spikes (purple). Figure adapted from [4].

The ChiroVox Dataset and Bat Echolocation Calls

Echolocation is a species-specific auditory trait of *Rhinolophus affinis* and *Rhinolophus pearsonii*. As a mammal with a highly specialized auditory system independently evolved from humans, bats offer a compelling test of whether sparse coding is a universal principle of mammalian auditory processing.

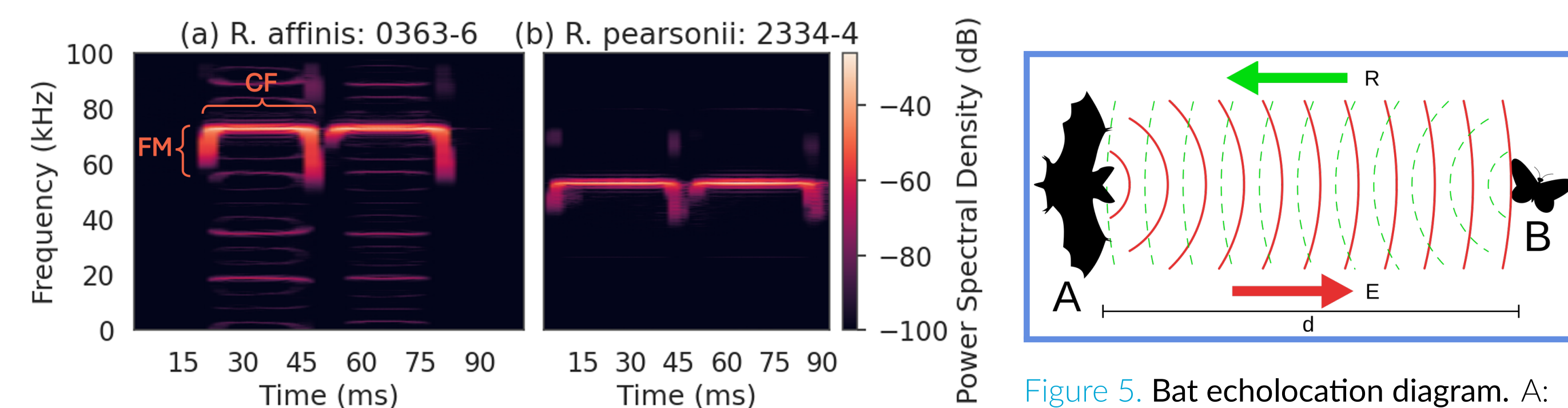


Figure 4. Echolocation call spectrograms of two Rhinolophidae species. (a) *R. affinis* calls at 72.9 kHz, with visible secondary harmonics (faint horizontal lines). (b) *R. pearsonii* calls at 53.2 kHz. [5]

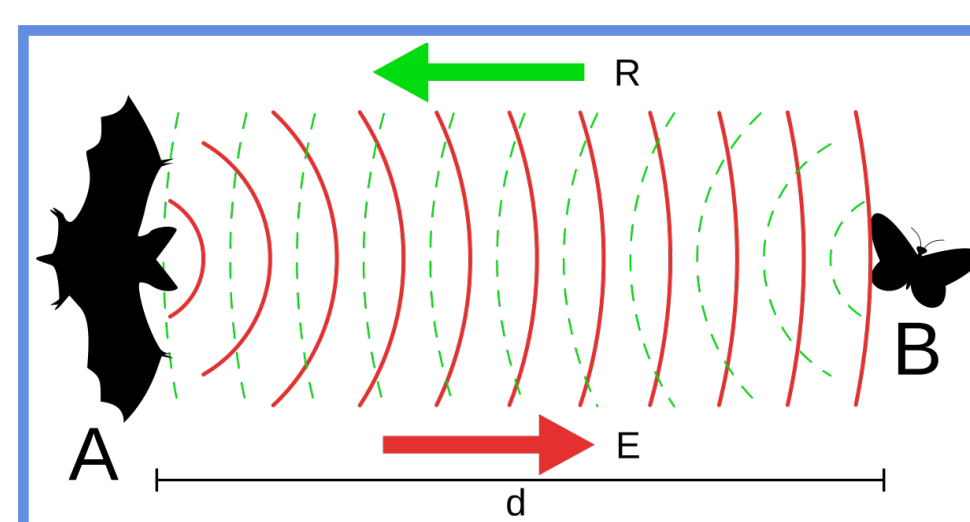


Figure 5. Bat echolocation diagram. A: Bat, B: Prey, E: Echolocation call emitted by the bat, R: Echo response received by the bat.

Research Questions Testing the Efficient Coding Hypothesis

RQ 1. Characteristics: What **spectral characteristics** define the structure of the kernels learned through the sparse coding of bat vocalizations?

RQ 2. Efficiency: To what extent do sparse representations achieve **greater coding efficiency** compared to traditional signal representation methods, such as Fourier and wavelets?

RQ 3. Specialization: To what degree do the learned kernels show **functional specialization**, with clusters of similar activation profiles encoding specific variations in bat calls?

RQ 4. Sparsity: To what extent do the learned representations exhibit **sparsity**, with a high prevalence of inactive (near-zero) coefficients across the kernel dictionary?

Frequency Distribution Analysis and Denoising

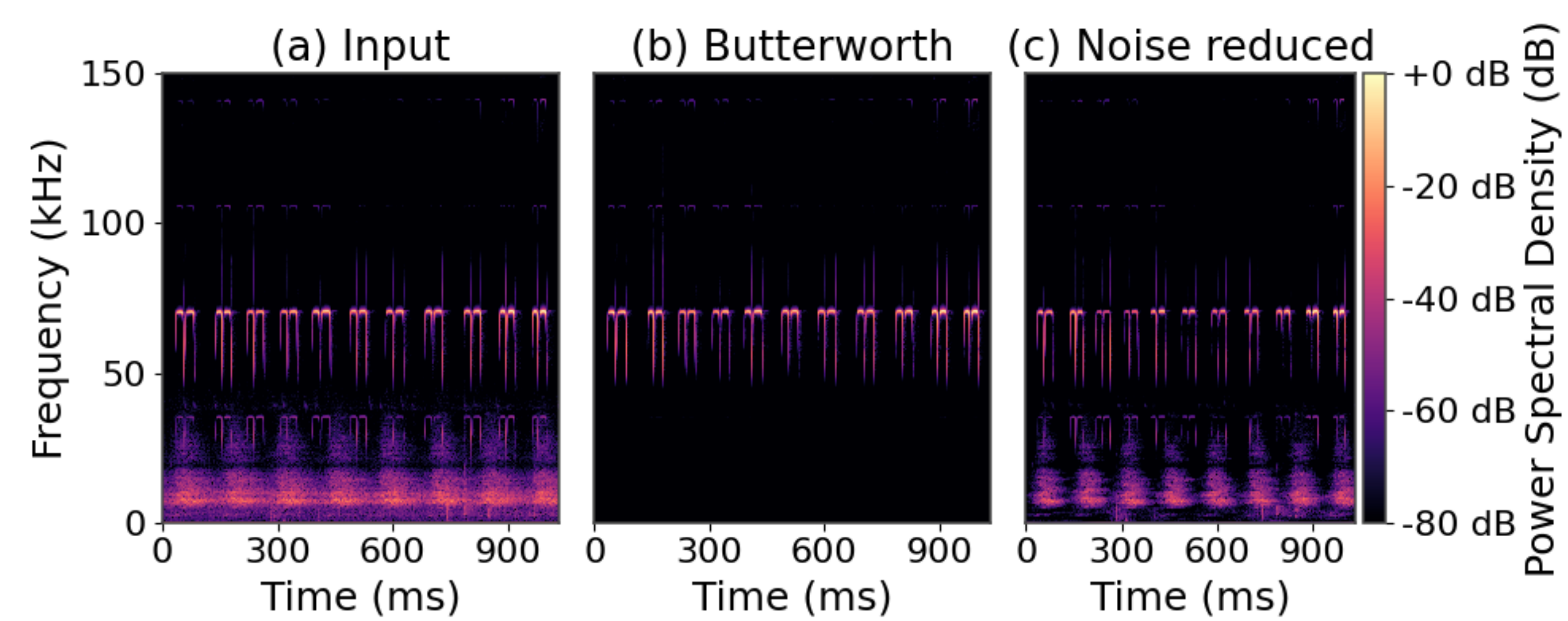


Figure 6. Denoising performance on *R. affinis* echolocation call. Recording with 500 kHz sample rate, spectral peak at 70.56 kHz, and Butterworth cutoff at 49.39 kHz.

Energy-Based Bat Call Detection

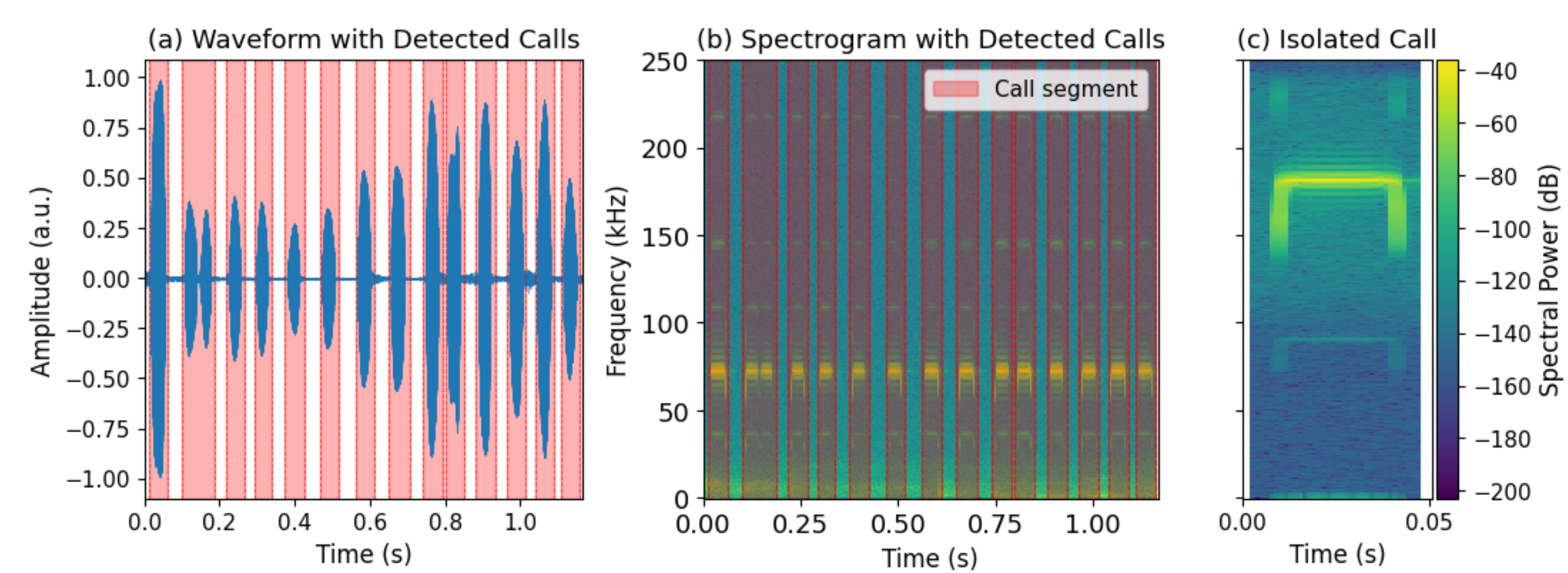


Figure 7. Energy-based call detection for *R. affinis*. Calls are identified where energy exceeds a 3 dB threshold and padded with 5 ms to make sure to capture FM sweeps. [6]

References

- [1] M. S. Lewicki, "Efficient coding of natural sounds," *Nature Neuroscience*, vol. 5, pp. 356–363, 2002.
- [2] E. C. Smith and M. S. Lewicki, "Efficient auditory coding," *Nature*, 2006.
- [3] S. G. Mallat and Z. Zhang, "Matching pursuits with time-frequency dictionaries," *IEEE Transactions on Signal Processing*, vol. 41, no. 12, pp. 3397–3415, 1993.
- [4] Wikipedia contributors, "Matching pursuit," 2026, accessed: 2026. [Online]. Available: https://en.wikipedia.org/wiki/Matching_pursuit
- [5] T. Görföl, J. C.-C. Huang, G. Csorba, D. Gyfirössy, P. Estók, T. Kingston, and et al., "Chirovox: a public library of bat calls," *PeerJ*, vol. 10, p. e12445, 2022.
- [6] M. D. Skowronski and M. B. Fenton, "Quantifying bat call detection performance of humans and machines," *Journal of the Acoustical Society of America*, vol. 125, no. 1, pp. 513–521, 2009.

Spectral Characteristics (RQ 1) and Efficiency (RQ 2)

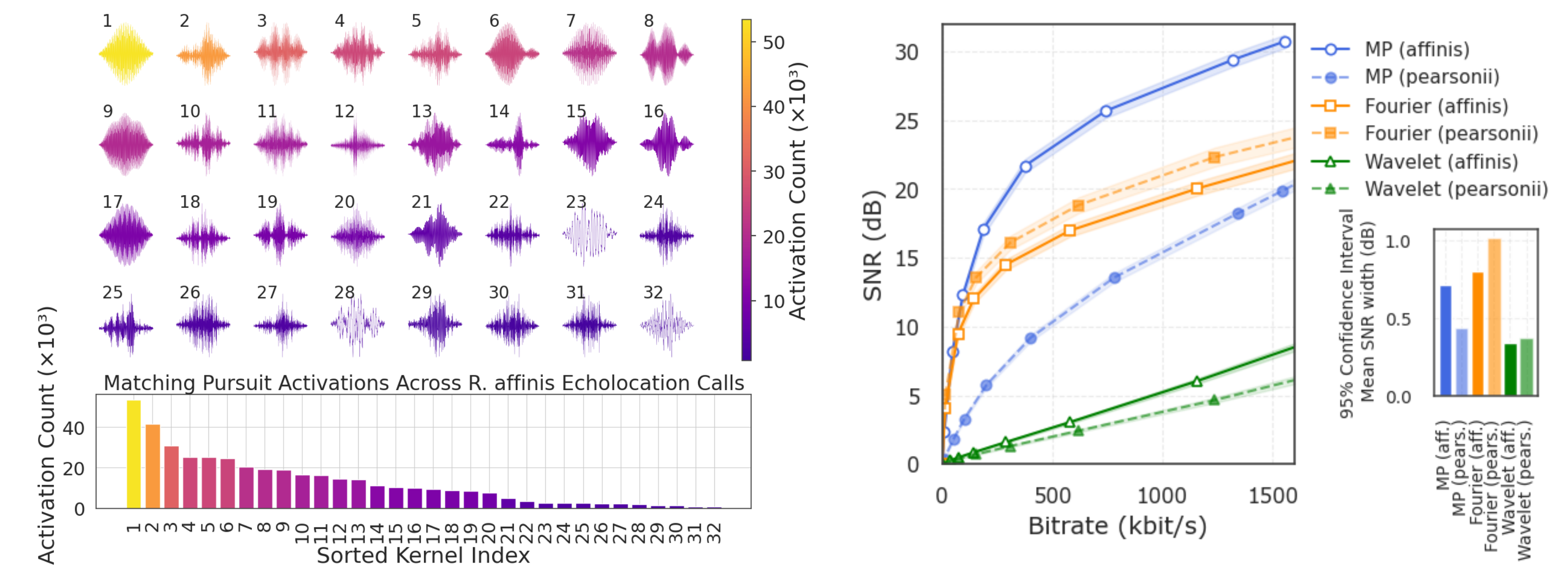


Figure 8. Kernel activation in *R. affinis* echolocation. Dictionary sorted descending by number of activations in 70-spike reconstruction.

Figure 9. Fidelity rate of Matching Pursuit, Fourier, and wavelet coding for *R. affinis* and *R. pearsonii* calls. Shaded regions denote tight (<1 dB) 95% confidence intervals.

Functional Specialization (RQ 3)

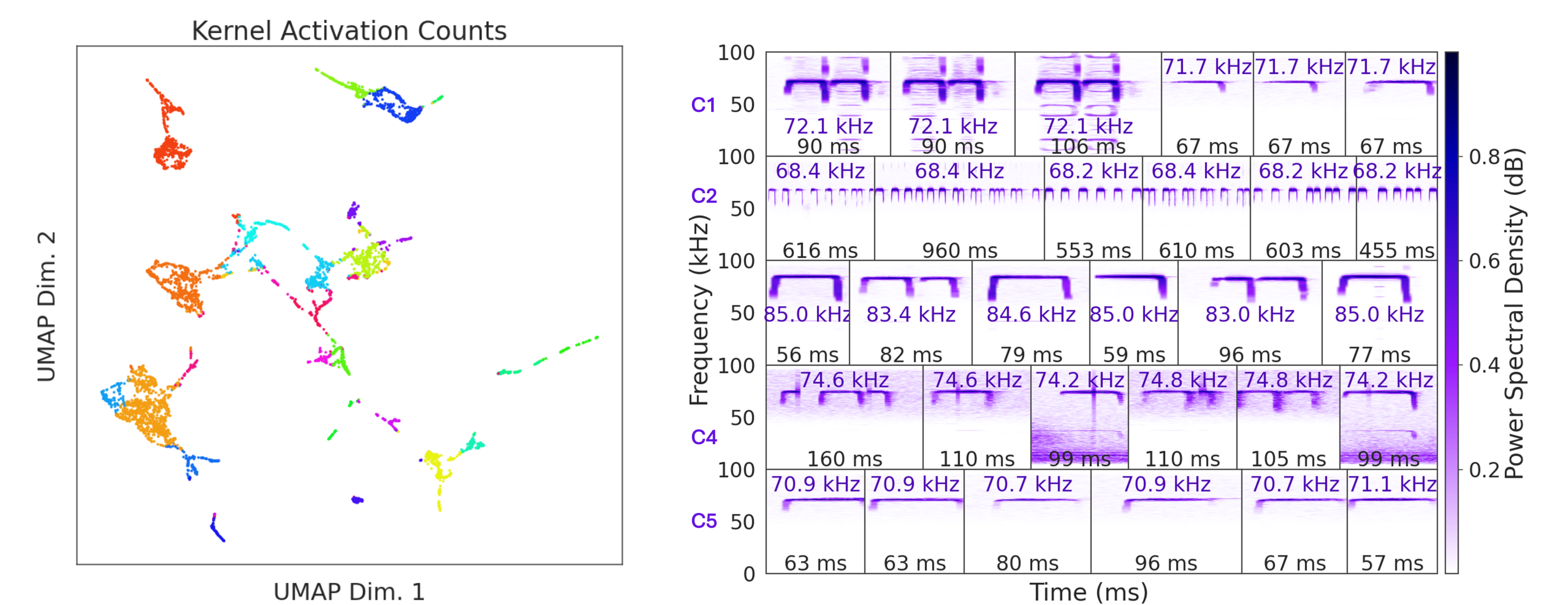


Figure 10. UMAP projection of bat call clusters. Each reconstruction represented by its kernel activation vector and clustered using KMeans with $k = 27$.

Figure 11. Auditory kernel-based clustering captures call diversity. Five selected clusters (C1–C5) with scaled durations.

Activation Sparsity (RQ 4)

Metric	200	2400
Gini (aff.)	0.984 ± 0.008	0.997 ± 0.001
Gini (pea.)	0.994 ± 0.004	0.998 ± 0.001
Hoyer (aff.)	0.959 ± 0.026	0.979 ± 0.011
Hoyer (pea.)	0.994 ± 0.014	0.990 ± 0.006
PQ (aff.)	$(3.59 \pm 0.28) \times 10^{-3}$	$(2.5 \pm 1.5) \times 10^{-4}$
PQ (pea.)	$(8.0 \pm 1.6) \times 10^{-4}$	$(1.2 \pm 0.8) \times 10^{-4}$

Table 1. Sparsity metrics. Mean $\pm$ SD of Gini, Hoyer, and PQ show high sparsity, with *R. pearsonii* sparser than *R. affinis*.

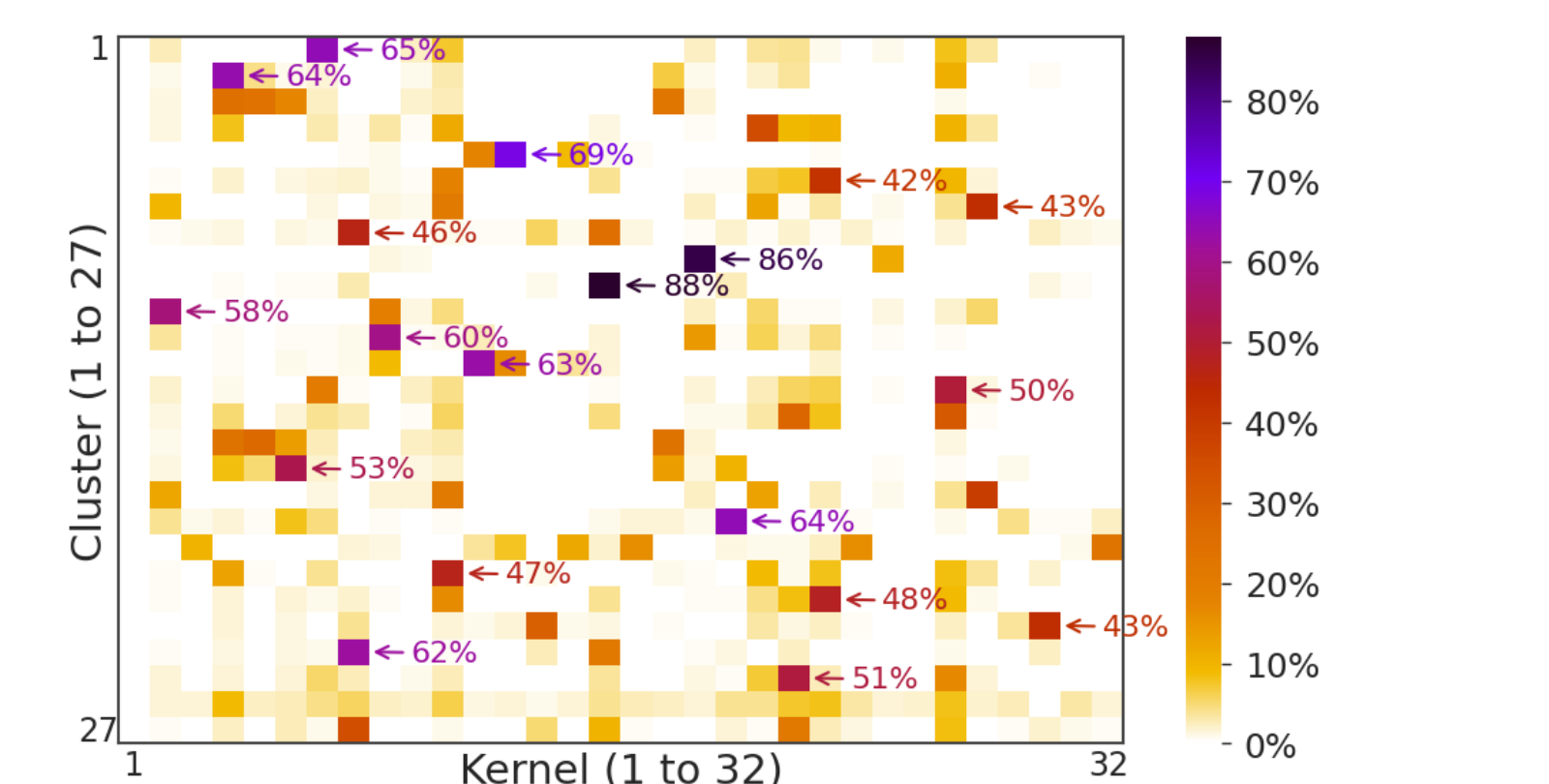


Figure 12. Cluster sparsity of kernel activation distribution. Contributions >40% are labeled. Mean cluster Gini = 0.8745.

Conclusion

The learned kernels are sparse, functionally specialized, and structured to reflect the acoustic properties of bat calls, consistent with efficient coding principles. This suggests that sparse coding has biological relevance beyond human speech and can provide interpretable computational foundation for analyzing and synthesizing animal vocalizations, supporting future research in decoding animal sounds and interspecies communication.